

Harvesting the Volatility Risk Premium on Leveraged ETFs

with a Drawdown-Conditioned LEAP Tail Overlay

An Integrated Short-Vol Carry and Long-Vol Convexity Framework

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Abstract

This paper develops a unified options framework that combines two structurally opposed sources of edge into a single portfolio architecture deployed on triple-leveraged equity ETFs (TQQQ, UPRO). The first component is a systematic short-volatility carry engine that harvests the volatility risk premium (VRP) through the cash-secured put / covered call “wheel” on the underlying ETF. The second is a drawdown-conditioned long-volatility convexity overlay implemented through long-dated equity anticipation securities (LEAPS) purchased mechanically when the underlying index drawdown from all-time-high exceeds a threshold D^* and implied volatility trades below trailing realized volatility. We formalize the VRP as the spread between option-implied and subsequently realized volatility, building on the foundational results of Coval and Shumway (2001) and Bakshi and Kapadia (2003), and we show that this premium is regime-dependent: it compresses sharply at the front of the term structure during persistent uptrends, while remaining materially positive in the 90–120 days-to-expiry (DTE) bucket. We model the VRP term structure as a regime-switching Ornstein–Uhlenbeck process per DTE bucket and derive the half-life of mean reversion as the operative quantity governing tenor selection. We then characterize the LEAP overlay as a stacked-convexity trade combining the daily-reset $3\times$ ETF structure, option delta/gamma expansion, and the conditional expected recovery move from extreme drawdown. The two components are structurally negatively correlated in their edge profiles: the wheel bleeds in exactly the regime where the LEAP trigger fires. We provide the academic basis for the joint edge, the practical implementation rules, and the risk overlays — vol-of-vol, beta slippage, path dependency, and pattern-day-trader constraints — required for institutional-grade deployment.

Keywords: volatility risk premium; leveraged ETFs; covered call wheel; LEAPS; convexity; tail risk; term structure; regime-switching; Ornstein–Uhlenbeck.

JEL Classification: G11, G12, G13, G14.

1. Introduction

The volatility risk premium (VRP) — the systematic wedge between option-implied volatility under the risk-neutral measure and subsequently realized volatility under the physical measure — is one of the most robust and academically documented compensations available to capital providers in derivatives markets. Coval and Shumway (2001) first demonstrated that delta-hedged S&P 500

option positions earn negative average returns, implying that long-volatility positions systematically lose money and that volatility sellers are compensated for absorbing volatility risk from hedgers. Bakshi and Kapadia (2003) sharpened this result by isolating the volatility risk premium from directional exposure through delta-neutral straddles. Carr and Wu (2009) decomposed the premium into pure variance, jump, and convexity components, providing the modern decomposition that underpins the variance-swap industry.

Despite the academic consensus on the existence of a structural VRP, two persistent challenges complicate its practical extraction. First, the premium is *not constant across the term structure* and *not constant across regimes*. Short-dated implied volatility compresses sharply toward realized volatility during strong directional trends, eroding the harvestable spread at the front of the curve precisely when systematic option sellers are most exposed. Second, the strategies that harvest the VRP earn their carry by accepting the left-tail risk that the premium exists to compensate. In a severe drawdown, the same regime that delivers months of patient theta accrual can be undone in a handful of days.

This paper develops an integrated framework that addresses both challenges by combining two structurally opposed sources of edge. The carry component is a cash-secured-put / covered-call wheel deployed on triple-leveraged equity ETFs (TQQQ, UPRO), where the multiplied implied volatility on the underlying ETF inflates the per-contract premium without proportionally inflating the structural cost of running the strategy. The tail-protection component is a long-dated equity anticipation security (LEAP) overlay triggered mechanically when the underlying index breaches a fixed drawdown threshold D^* from all-time-high and implied volatility trades below trailing realized volatility — a configuration that historically marks the back end of liquidation cascades and the front end of recovery rallies on leveraged equity ETFs.

The architectural insight is that these two components are *structurally negatively correlated* in their edge profiles. The wheel earns carry during quiet, trending, mean-reverting regimes and bleeds during volatility shocks. The LEAP earns its return from a stacked-convexity payoff in exactly the regime that hurts the wheel. The combined portfolio does not stack two correlated bets on the same underlying premium — it constructs a basis that spans short-vol carry on one axis and long-vol convexity on the other.

We organize the paper as follows. Section 2 formalizes the volatility risk premium and its empirical structure. Section 3 reviews the path-dependency, beta slippage, and vol-drag properties of triple-leveraged ETFs that make them the natural vehicle for VRP harvesting. Section 4 specifies the wheel as a discrete-time short-volatility strategy and derives the relevant Greeks. Section 5 models the VRP term structure as a regime-switching Ornstein–Uhlenbeck process per DTE bucket. Section 6 formalizes the drawdown-conditioned LEAP overlay and decomposes its three layers of convexity. Section 7 characterizes the combined portfolio and the structural negative correlation between its components. Section 8 addresses the risk overlays and operational constraints. Section 9 concludes.

2. The Volatility Risk Premium

2.1. Definition Under Both Measures

Let $\sigma_{IV}(t, T)$ denote the option-implied volatility at time t for horizon T , extracted from the at-the-money option chain (or, more rigorously, as the model-free implied volatility via the Bakshi–Kapadia–Madan (2003) replication). Let $\sigma_{RV}(t, T)$ denote the volatility realized over $[t, T]$ under

the physical measure $\mathbb{P}$, computed as the annualized square root of the sum of squared log-returns:

$$\sigma_{RV}(t, T) = \sqrt{\frac{1}{T-t} \sum_{u=t}^T r_u^2}. \quad (1)$$

The volatility risk premium for tenor $T - t$ is the expected spread between the implied and subsequently realized volatility:

$$\text{VRP}(t, T) \equiv \mathbb{E}^{\mathbb{Q}}[\sigma_{IV}(t, T)] - \mathbb{E}^{\mathbb{P}}[\sigma_{RV}(t, T)]. \quad (2)$$

Empirically, $\text{VRP}(t, T) > 0$ on average across asset classes and tenors. This is the structural premium harvested by volatility sellers. The same construct can be expressed in variance space as $\mathbb{E}^{\mathbb{Q}}[\sigma^2] - \mathbb{E}^{\mathbb{P}}[\sigma^2]$, which is the form replicated exactly by the fixed leg of a variance swap (Demeterfi et al., 1999); we work in volatility space because it is the natural unit for option-pricing in practice and avoids the Jensen wedge that complicates the variance-space interpretation.

2.2. Three-Component Decomposition

Following the decomposition surveyed in Bakshi and Kapadia (2003) and Carr and Wu (2009), the VRP can be partitioned into three components:

$$\text{VRP} = \underbrace{\pi_{\text{ins}}}_{\text{insurance}} + \underbrace{\pi_{\text{jump}}}_{\text{jump risk}} + \underbrace{\pi_{\text{conv}}}_{\text{convexity}}. \quad (3)$$

The insurance component compensates option sellers for absorbing hedging demand from institutional buyers. The jump component prices left-tail discontinuities that the diffusive Black–Scholes–Merton kernel cannot capture. The convexity component arises because variance is itself a convex payoff, generating a Jensen’s-inequality wedge between $\mathbb{E}^{\mathbb{Q}}[\sigma^2]$ and $(\mathbb{E}^{\mathbb{Q}}[\sigma])^2$. All three components reward the seller in normal regimes and punish the seller in shock regimes.

2.3. Why the VRP Exists

The VRP is not an arbitrage. It is a risk premium paid by hedgers, who systematically demand protection against equity-market drawdowns, to speculators, who provide that protection by selling volatility. The premium is positive on average because the demand for protection is structurally asymmetric: pension funds, insurance companies, and structured-product issuers are natural buyers of volatility, while the population of natural sellers — well-capitalized speculators willing to absorb left-tail risk — is smaller. The premium therefore reflects the marginal cost of risk-bearing capacity in volatility, not a mispricing.

3. Leveraged ETFs as the Harvesting Vehicle

3.1. The Daily Reset Mechanic

A k -times leveraged ETF (e.g., $k = 3$ for TQQQ on the NASDAQ-100, UPRO on the S&P 500) targets a daily return equal to k times the daily return of the underlying index:

$$r_t^{\text{LETF}} = k \cdot r_t^{\text{index}}. \quad (4)$$

Because the leverage ratio is reset every trading day, the multi-day compounded return is path-dependent. For n trading days:

$$R_{0,n}^{\text{LETF}} = \prod_{t=1}^n (1 + k \cdot r_t^{\text{index}}) - 1 \neq k \cdot R_{0,n}^{\text{index}}. \quad (5)$$

The wedge between the leveraged-ETF compounded return and k times the underlying's compounded return is the well-documented *beta slippage* (Cheng and Madhavan, 2009; Avellaneda and Zhang, 2010). For a log-normal index with volatility σ and zero drift, the expected multi-day slippage is approximately:

$$\mathbb{E} [R_{0,n}^{\text{LETF}}] - k \cdot \mathbb{E} [R_{0,n}^{\text{index}}] \approx -\frac{k(k-1)}{2} \sigma^2 \cdot n. \quad (6)$$

This is the volatility drag: a deterministic, quadratic penalty that scales with $k(k-1)$, with realized variance, and with holding period.

3.2. Implications for Option Premium

The slippage in equation (6) is the structural cost of holding leveraged ETFs through chop. But the same daily reset that creates vol drag also inflates the realized and implied volatility of the leveraged ETF relative to the underlying, since:

$$\sigma^{\text{LETF}} \approx k \cdot \sigma^{\text{index}}, \quad (7)$$

which means that for a $3\times$ ETF, the at-the-money option premium is approximately tripled in dollar terms per unit of notional. The wheel strategy converts this inflated premium into carry. The vol drag becomes the structural cost being financed by the VRP: a continuous short-vol carry stream offsets the path-dependency penalty of holding the underlying leveraged ETF, and the residual is the strategy's edge.

3.3. Why $3\times$ Rather Than $1\times$

A wheel deployed on a $1\times$ ETF (e.g., SPY) earns the VRP on the underlying. A wheel deployed on a $3\times$ ETF earns approximately the same fractional VRP applied to a tripled-volatility underlying, generating approximately three times the absolute dollar premium per contract. Because the operational cost of running the wheel (commissions, slippage, capital allocation, monitoring overhead) does not scale with k , the after-cost yield on the leveraged version is materially higher. The structural risk — a path-dependent drawdown during a sustained vol regime — is what the LEAP overlay in Section 6 is designed to hedge.

4. The Wheel as a Discrete-Time Short-Volatility Strategy

4.1. The Cycle

The wheel is a continuous-rolling sequence of two short-option positions:

1. **Cash-secured put (CSP).** Sell an out-of-the-money put with strike $K_p < S_t$ and tenor τ . Hold cash equal to $100 \cdot K_p$ as collateral. If the put expires worthless, collect premium P_p and repeat. If assigned, take delivery of 100 shares at K_p .

2. **Covered call (CC).** Once assigned, sell an out-of-the-money call with strike $K_c > S_t$ and tenor τ . If the call expires worthless, collect P_c and repeat. If assigned, deliver 100 shares at K_c and return to step 1.

The wheel is a continuous short-gamma, short-vega, long-theta position with discrete reset points at each expiry.

4.2. Entry Discipline

The empirically robust entry rule is to sell options at the $\Delta \in [0.20, 0.30]$ band measured at trade initiation. This corresponds to strikes approximately 1.0–1.5 standard deviations out-of-the-money under the implied distribution. The choice trades off three considerations:

1. Lower delta $\Rightarrow$ higher win rate but smaller premium per trade.
2. Higher delta $\Rightarrow$ larger premium but higher assignment probability and faster path into the covered-call leg.
3. The 0.20–0.30 band maximizes the product of premium per trade and survival probability over multi-cycle horizons.

4.3. The Greek Profile

At trade initiation, the short put has:

$$\Delta_{\text{CSP}} \in [-0.30, -0.20], \quad \Gamma_{\text{CSP}} > 0, \quad \mathcal{V}_{\text{CSP}} < 0, \quad \Theta_{\text{CSP}} > 0. \quad (8)$$

The strategy's edge over time is captured by the theta integral:

$$\Pi_{0,T}^{\text{wheel}} = \int_0^T \Theta_u du - \int_0^T \mathcal{V}_u d\sigma_u - \frac{1}{2} \int_0^T \Gamma_u d\langle S \rangle_u, \quad (9)$$

where the first term is the theta carry, the second is the vega mark-to-market exposure, and the third is the gamma exposure to realized variance (the quadratic variation of the underlying). In expectation, the theta carry dominates if and only if implied volatility exceeds realized volatility over $[0, T]$ — i.e., if and only if the VRP is positive. This is the precise sense in which the wheel is a discrete-time VRP harvester.

4.4. Collateral Yield as the Performance Metric

Because the wheel ties up cash collateral equal to $100 \cdot K_p$ on the put leg, the natural performance metric is not raw P&L but annualized yield on collateral:

$$y_\tau^{\text{coll}} = \frac{P_p}{100 \cdot K_p} \cdot \frac{365}{\tau}. \quad (10)$$

This converts heterogeneous tenor selections onto a common axis and is the right object to track over multi-cycle horizons.

5. The VRP Term Structure and the Tenor-Selection Problem

5.1. Why Tenor Matters

The wheel is implemented at a specific tenor. The classical practitioner default is 30–45 DTE, motivated by the rapid theta decay that occurs in the final month before expiry. But the harvestable VRP is not constant across the term structure. In persistent uptrending regimes — exactly the regimes that produced the post-2023 leveraged-ETF rally — the front of the IV curve compresses faster than the back, because:

1. Realized variance falls toward zero in trending regimes.
2. Short-dated implied variance tracks realized variance with a short lag.
3. Long-dated implied variance prices in a mean-reversion expectation that the trend will not persist indefinitely.

The result is a regime-dependent VRP term structure: $\text{VRP}(t, t + \tau)$ collapses for short τ but remains positive for $\tau \in [90, 120]$ days.

5.2. An Ornstein–Uhlenbeck Model of the VRP Per DTE Bucket

Let X_t^τ denote the VRP at time t for tenor bucket τ (e.g., $\tau \in \{30, 60, 90, 120\}$ days). We model X_t^τ as a mean-reverting Ornstein–Uhlenbeck process:

$$dX_t^\tau = \theta_\tau (\mu_\tau - X_t^\tau) dt + \sigma_\tau dW_t. \quad (11)$$

The discrete-time AR(1) representation is the operative form for estimation:

$$X_{t+1}^\tau = X_t^\tau e^{-\theta_\tau \Delta t} + \mu_\tau (1 - e^{-\theta_\tau \Delta t}) + \epsilon_t. \quad (12)$$

Estimating via the regression $X_{t+1}^\tau = \alpha + \beta X_t^\tau + \epsilon_t$ yields:

$$\theta_\tau = -\frac{\ln \beta}{\Delta t}, \quad \mu_\tau = \frac{\alpha}{1 - \beta}, \quad \sigma_\tau = \text{sd}(\epsilon) \sqrt{\frac{-2 \ln \beta}{\Delta t(1 - \beta^2)}}. \quad (13)$$

The actionable summary statistic is the *half-life* of mean reversion:

$$\text{HL}_\tau = \frac{\ln 2}{\theta_\tau}. \quad (14)$$

Empirically, $\text{HL}_{30} \ll \text{HL}_{90} < \text{HL}_{120}$: short-dated VRP reverts on a horizon of days, long-dated VRP on a horizon of weeks or months. The longer half-life at the back of the curve means that the back end spends more time above its long-run mean in any given regime — exactly the property that the tenor-migration argument requires.

5.3. Regime Conditioning

A homogeneous OU with constant μ_τ misses the regime dependence that motivates the tenor-selection problem. Two extensions address this:

Regime-switching OU. Fit two parameter sets conditional on a regime indicator $Z_t \in \{\text{trend}, \text{chop}\}$. The trend regime is operationalized as “index above the 50-day moving average with positive slope.”

Empirically, $\mu_{30}^{\text{trend}} < \mu_{30}^{\text{chop}}$ while $\mu_{90}^{\text{trend}} \approx \mu_{90}^{\text{chop}}$ — confirming that the front of the curve is regime-sensitive and the back is regime-robust.

OU with exogenous drift. A more parsimonious specification:

$$dX_t^\tau = \theta_\tau (\mu_\tau + \gamma_\tau Z_t - X_t^\tau) dt + \sigma_\tau dW_t, \quad (15)$$

where γ_τ measures the regime sensitivity of the equilibrium. The hypothesis $\gamma_{30} \ll \gamma_{90}$ is the formal version of the tenor-migration claim.

5.4. The Tenor-Selection Decision Rule

Conditional on the OU estimates, the tenor-selection rule is:

1. Compute the current z -score $z_t^\tau = (X_t^\tau - \mu_\tau)/(\sigma_\tau/\sqrt{2\theta_\tau})$ for each DTE bucket.
2. Select the tenor with the highest z_t^τ above a threshold z^* (e.g., $z^* = 0$).
3. If $z_t^{30} < 0$ but $z_t^{90} > 0$, migrate to the 90-DTE bucket; the short end is selling cheap premium and the back end is selling rich premium.
4. Re-evaluate at each cycle reset.

This converts the tenor decision from a static practitioner default into a state-dependent allocation that follows the harvestable premium along the term structure.

5.5. Caveats

OU assumes Gaussian innovations. Empirical VRP has fat left tails: vol spikes blow out the spread negatively in episodic bursts (Aug 2015, Feb 2018, Mar 2020). For risk management, the z -score signal should be truncated rather than scaled symmetrically; extreme negative z is more often the front of a regime break than a reversion opportunity. The OU framework provides the equilibrium structure; the LEAP overlay in Section 6 handles the residual jump risk that the OU model cannot absorb.

6. The Drawdown-Conditioned LEAP Overlay

6.1. The Trigger

The LEAP overlay is a binary, mechanical signal that converts a continuous market state into a discrete trade. The trigger fires when both conditions hold simultaneously:

$$\mathbb{1}_{\text{LEAP}}(t) = \mathbb{1}\{D_t \geq D^*\} \cdot \mathbb{1}\{\sigma_t^{\text{IV}} < \sigma_t^{\text{RV}}\}, \quad (16)$$

where D_t is the drawdown of the underlying index from its all-time high, D^* is the threshold (calibrated at 35%), σ_t^{IV} is the prevailing implied volatility on the leveraged ETF, and σ_t^{RV} is the trailing realized volatility.

The drawdown threshold isolates the rare regime in which the index has already absorbed a major liquidation. The IV/RV condition filters for the subsequent state in which implied volatility has begun to compress below realized volatility — a configuration that historically marks the back end of the panic phase and the front end of the recovery rally. The trigger is therefore not a market-timing rule on the underlying; it is a state-detection rule on the joint dynamics of price and volatility.

6.2. The Instrument

Conditional on the trigger, the trade is a long position in a 2-year LEAP call on the leveraged ETF (UPRO), purchased deep in-the-money or near-the-money to maximize the delta sensitivity to the conditional expected recovery move while preserving the convexity gain from gamma expansion as the underlying rallies.

6.3. Three Layers of Stacked Convexity

The expected return on the trigger-conditional trade is not a single risk premium but a product of three:

$$\mathbb{E} [R^{\text{LEAP}} \mid \mathbb{1}_{\text{LEAP}} = 1] = \underbrace{f_1(R^{\text{index}})}_{3 \times \text{ETF}} \cdot \underbrace{f_2(\Delta, \Gamma)}_{\text{option Greeks}} \cdot \underbrace{f_3(D^*, \text{horizon})}_{\text{conditional recovery}}. \quad (17)$$

The first factor is the deterministic $k = 3$ leverage of the underlying ETF on the index recovery. The second factor is the convex payoff of the option as the underlying rallies, captured by the delta as it approaches one and the gamma as the strike is breached. The third factor is the conditional expected recovery from extreme drawdown, which empirically is positive and economically large at the 35% threshold based on the historical distribution of S&P 500 drawdowns since 1928.

6.4. Why a Full Ride Rather Than a Vol-Triggered Exit

A naive exit rule would liquidate the LEAP when implied volatility compresses back to a normal regime, on the grounds that vega bleed begins to dominate. We argue against this rule for three reasons:

1. During the recovery, the delta of an in-the-money 2-year LEAP rises toward one. Delta gain from the underlying rally dominates vega loss from IV compression by an order of magnitude during the first 50% of the recovery.
2. Gamma expansion is greatest in the first 30% of the recovery move; harvesting this requires holding through the IV-compression period.
3. The 2-year tenor was chosen precisely to insulate the trade from short-horizon vega oscillation. Exiting on vega compression abandons the structural reason for choosing LEAPS over shorter-dated calls.

6.5. Why a Mechanical Trigger Rather Than Discretion

The decision to enter at $D^* = 35\%$ is mechanical by design. Discretionary judgment about “regime” is the primary source of behavioral failure in tail-trade strategies: by the time the discretionary trader is convinced the regime has bottomed, the recovery is well underway and the convexity has compressed. The mechanical trigger replaces the regime-classification problem with a single observable variable (drawdown) and a single state filter (IV/RV), both of which are immune to narrative drift. This is the same logic that motivates rules-based momentum and trend-following strategies, applied here to the convexity domain.

7. The Integrated Portfolio

7.1. Structural Negative Correlation

The wheel and the LEAP overlay are negatively correlated in their edge profiles by construction. Table 1 summarizes the structural exposures.

Table 1: Edge-profile decomposition of the two components.

Dimension	Wheel (CSP/CC)	LEAP overlay
Vega exposure	Short	Long
Gamma exposure	Short	Long
Theta exposure	Long (positive carry)	Short (cost)
Best regime	Quiet, trending	Post-drawdown
Worst regime	Vol shock / drawdown	Quiet trending
Edge source	VRP (short volatility)	Conditional convexity
Premium direction	Receive	Pay
Trigger	Continuous	Conditional ($D \geq D^*$)

7.2. Capital Allocation

The portfolio holds three sleeves:

1. Wheel collateral: capital reserved against assigned CSPs at each cycle.
2. Cash reserve: approximately 25% of the account held in T-bills as dry powder, used to fund the LEAP overlay when triggered.
3. Triggered LEAP: zero until $\mathbb{K}_{\text{LEAP}} = 1$, then a discretized allocation from the cash reserve.

The 25% cash reserve is the operational solution to the LEAP-funding problem. It carries an opportunity cost during long uneventful regimes — the same regimes that maximize the wheel’s collateral yield — but converts the LEAP from a leveraged optional addition into a reliably fundable mechanical trade. The carry foregone on the reserve is the explicit insurance premium paid for the optionality.

7.3. The Combined Return Profile

Let R_t^W denote the wheel return and R_t^L denote the LEAP-overlay return per unit of allocated capital. The portfolio return is:

$$R_t^{\text{port}} = w \cdot R_t^W + (1 - w) \cdot \mathbb{K}_{\text{LEAP}}(t) \cdot R_t^L + (1 - w) \cdot (1 - \mathbb{K}_{\text{LEAP}}(t)) \cdot r_t^f, \quad (18)$$

where w is the wheel allocation, $1 - w$ is the cash-plus-LEAP sleeve, and r_t^f is the risk-free rate earned on the reserve when no LEAP is held. Because $\text{Cov}(R^W, \mathbb{K}_{\text{LEAP}} \cdot R^L) < 0$ by construction — the LEAP fires when the wheel is bleeding — the combined variance is strictly less than the variance-weighted sum of the components.

8. Risk Overlays and Operational Constraints

8.1. Path-Dependency Risk

The single largest source of unrecoverable loss is a sustained vol regime that holds the leveraged ETF in chop for an extended period. Equation (6) quantifies the deterministic vol-drag cost; the wheel carry must exceed this drag in expectation, and the LEAP overlay must be available to recover convexity when the chop resolves into a drawdown. The system is designed to be solvent through arbitrary chop horizons given (i) the cash reserve, (ii) the conditional rather than continuous LEAP allocation, and (iii) the entry-discipline constraint that maintains assignment probability below a fixed bound.

8.2. Pattern Day Trader Constraint

Under U.S. broker rules, accounts below \$25,000 in equity are restricted from executing more than three day trades in a rolling five-day window. The wheel strategy is structurally not a day-trading strategy — its cycles measure in weeks — but the LEAP overlay’s optional intra-cycle adjustments (rolling, partial closes) require monitoring of the PDT constraint at account sizes near the threshold. Above \$25,000, the constraint is not binding.

8.3. Vol-of-Vol Risk

The OU model in equation (11) assumes the diffusion coefficient σ_τ is constant. Empirically, vol-of-vol is itself time-varying and correlates positively with regime stress. During stress regimes, the z -score signal becomes unreliable because the stationary variance of OU is itself nonstationary. The operational response is to gate trade entries on a regime indicator: do not enter new wheel cycles when vol-of-vol exceeds a fixed percentile, and instead let the existing positions roll off into the cash reserve.

8.4. Assignment and Tax Treatment

The wheel generates a mix of short-term capital gains (from premium received on options held to expiry or closed within one year) and potentially long-term gains (from shares assigned and held more than one year before the covered-call assignment). The 2-year LEAP, if held to expiry, qualifies for long-term capital-gains treatment on the option itself. The portfolio’s after-tax yield depends on the holding-period composition; the deployment cadence should be planned in coordination with the investor’s tax horizon.

9. Conclusion

This paper has developed an integrated derivatives framework that combines two structurally opposed sources of edge on triple-leveraged equity ETFs. The carry component — a cash-secured-put / covered-call wheel deployed at the 0.20–0.30 delta band — harvests the volatility risk premium in continuous-rolling form. The tail-protection component — a 2-year LEAP triggered mechanically when the index drawdown exceeds 35% from all-time-high and implied volatility trades below trailing realized volatility — captures three stacked layers of convexity (leveraged ETF structure, option Greeks, conditional recovery) in exactly the regime where the carry component bleeds.

We have argued that the VRP is regime-dependent across the term structure, with short-dated premium collapsing in persistent uptrends while the 90–120 DTE bucket retains a positive and

economically meaningful spread. We have modeled the VRP per DTE bucket as a regime-switching Ornstein–Uhlenbeck process, derived the mean-reversion half-life as the operative quantity for tenor selection, and specified a state-dependent allocation rule that follows the harvestable premium along the curve. We have characterized the LEAP overlay as a mechanical trigger that replaces discretionary regime classification with two observable variables, and we have argued for a full-recovery hold rather than a vol-triggered exit on the grounds that delta gain dominates vega loss during the first half of the recovery.

The architectural insight is that the wheel and the LEAP are negatively correlated in their edge profiles by construction. The combined portfolio is not a stack of two correlated bets on the same underlying premium; it is a basis that spans short-vol carry on one axis and long-vol convexity on the other. The resulting profile is structurally lower-variance than either component alone, with persistent positive carry in normal regimes and an embedded mechanical response when the carry regime breaks.

Three extensions are natural. First, an empirical calibration of the regime-switching OU parameters on TQQQ and UPRO option chains over a multi-year window would convert the framework from a structural specification into a back-testable system. Second, the addition of a third orthogonal sleeve — a cross-sectional equity factor such as residual momentum — would extend the portfolio basis from two dimensions to three. Third, a formal optimization over the wheel-LEAP allocation w as a function of the regime-conditional Sharpe ratios would replace the heuristic 75/25 split with a state-dependent rule. We leave these to subsequent work.

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